
Assignment 1 - Getting Started with Machine Learning

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1 Abstract

In this assignment, we evaluated the performance of K-Nearest Neighbors (KNN) and Decision Trees (DT) from scratch on two benchmark datasets: the Heart Disease dataset(Binary Classification) and the Penguins dataset (multi-class classification). Our experiments involved analyzing accuracy, the impact of hyperparameters on the results, and feature importance. We found that on the Penguins dataset, both models performed similarly where they both showed high accuracies, with KNN achieving slightly higher. Conversely, on the Heart Disease dataset, the Decision Tree classifier outperformed KNN. These results highlight the importance of dataset characteristics in model selection and the importance of fine-tuning for optimal performance.

2 Introduction

Various machine learning classification algorithms exist in the world, each with their respective strengths and weaknesses. This study explores the implementation of two very popular ML algorithms, K-Nearest Neighbors (KNN) and Decision Trees (DT), and aims to compare the performance.

These models are applied to two distinct datasets: the Heart Disease dataset from the UCI Machine Learning Repository, and the Penguin dataset from Kaggle, both representing important data in the realm of health and biology. An important distinction to be made is that the Heart Disease dataset presented a binary classification task, predicting the presence or absence of heart disease, while the Penguin dataset involved a multi-class classification problem, identifying species of penguins into three categories.

While our most broad, overarching task was to implement both classification techniques on two different datasets, a large importance was placed on elevating this performance, in an attempt to get the highest test-accuracy possible for a given model on a given dataset. Performance was optimized in the following ways (more details in discussion).

1. Deliberately selecting features to remove from each dataset before training our models was very carefully considered, as classification models are greatly influenced by irrelevant features. According to prior research on the topic “Machine Learning Classifications of Coronary Artery Disease” [1] which used the same dataset as we did to determine heart disease, correctly selecting the features provided a significant improvement in all evaluation methods when compared to not doing so. (It is important to note that we used our own methods to remove these irrelevant features but cross-checked with this paper and arrived at the same conclusion of which to drop).
2. Implementing weighted KNN
3. Splitting the data into training, validation and testing data, and then using the validation set to select the best K and the best tree depth and evaluate the best choice on the test set.
4. Experimenting with different distance functions

Furthermore, here are techniques with which we experimented in order to improve the performance on the heart dataset, which had significantly worse performance than the penguin dataset:

- Splitting the dataset into 3: one with only categorical features, one with only numerical features, and one with mixed features. A lot of research exists that attempts to understand the role of mixed data on classification algorithms, to varying conclusions. For example, this research “The distance function effect on k-nearest neighbor classification for medical datasets” [2] reveals how different distance functions work better on different datasets. It concludes that euclidean distance works

better on only categorical and only numerical datasets, but underperforms on mixed datasets. Due to this uncertainty on what type of data works best, we thought it could be interesting to compare these 3 datasets and explore which one performed the best.

After conducting our analysis, we came to the following conclusions. In the heart data set, KNN with the parameter $k = 6$ was optimal and led to a test accuracy of 77%. The optimal depth of the DT was 3, which led to an accuracy of 88.33%. On the penguin dataset, KNN resulted in much higher percentages with the optimal $k=9$, we got the accuracy 99%. So did dt, with an optimal depth of 5 and test accuracy of 98.48%. For our heart dataset, the mixed data resulted in the highest accuracy.

In sum, by implementing these algorithms from scratch, this study offers a hands-on perspective on classical machine learning methods and their applications to real-world healthcare and biological datasets.

3 Methods

The K-Nearest Neighbors (KNN) algorithm is a simple yet effective non-parametric classification method. Given a new data point, KNN finds the K closest points in the training set based on a chosen distance function and assigns the most common label among them. The probability of a given class is computed as the fraction of the K neighbors that belong to that class. The choice of K significantly impacts the model's performance: a small K may lead to overfitting, while a large K can cause underfitting by smoothing decision boundaries excessively. [3]

To improve the performance of the model, we tried some approaches. First of all we implemented a model such that except for its default distance function (euclidean), it would be possible to try out different distance functions like manhattan. Also we ran the model with different K values to compare the model's performance in different variations and pick the best one so while trying to avoid underfitting data by growing the k but also not to choose a small value that would cause overfitting. The other option we added to our model was the weighted KNN. The idea is that not all k data points have the same influence on predicting the label for our point X , and logically closer training points should have more weight in the prediction. Finally we made sure to normalize the data points to avoid any issues regarding different features having different scaling.

These modifications allowed us to assess the sensitivity of KNN to different design choices and provided deeper insights into its performance across datasets with different type of characteristics. We built a decision tree and trained it using the recursive method "`_fit_tree()`". During the tree construction process, we used greedy search to select the best split features and split values. Then traverse all features and evaluate all possible split points for each feature. Next, we calculate the weighted cost of the divided subsets and select the optimal split point. The tree construction stops when the maximum depths are reached or the number of samples at the current node is less than `min_leaf_instances`.

To optimize the performance of decision trees, we tried three different cost functions: Misclassification rate, Entropy and Gini index.

When evaluating the models, we calculated the test accuracy and AUROC for the Heart Disease dataset and plotted ROC curves for comparison with KNN. For the Penguin dataset, we calculated the test accuracy and the average AUROC for all categories under different cost functions.

In addition, we calculated the top 5 most important features used by the decision tree and compared them with the important features selected by the mean difference method to explore the similarities and differences between different feature selection methods.

4 Datasets

The heart dataset started with 14 features, while the penguin one started with 6 features. To better understand the datasets, we performed exploratory data analysis and then performed certain necessary pre-processing steps. Our findings are as follows:

- The heart dataset is relatively balanced, with approximately equal numbers of positive and negative cases. The distribution was 53% for no heart disease and 46% positive for heart disease.
- For the penguin dataset, since the dataset ended up containing only four numerical features, it was well-suited for visualization. Using a 2D and 3D scatter plot, we could clearly see that the three species were well-separated in feature space, confirming that the dataset was highly suited for classification tasks.
- Cleaning the data involved removing missing rows, and, for the heart dataset, transforming a value of 1-4 for heart disease presence into a 1 in order to turn our target into a binary variable -1

(presence) or 0 (no presence). For the Penguin dataset, we first removed rows with missing values, including an entry where the sex column contained a "." instead of "Male" or "Female". Additionally, we dropped the "island" feature because it was not relevant for classification. Since all features except sex were numerical, we applied one-hot encoding with column dropping to avoid multicollinearity.

As mentioned previously, it was vital to drop the irrelevant features in order to optimize our dataset, as indicated by common knowledge about KNN and the research outlined in "Introduction". Here is how we determined which to drop:

- On the heart dataset, for numerical features, we calculated the difference of the mean of each feature in the positive group vs in the negative group to determine which features were irrelevant. For the categorical features, we began by visualizing their importance by plotting the percentage of positive points with that feature vs positive points without the same feature. In doing so, we got a sense of which features were potentially not important. For example, the percentage of data points with fbs (blood sugar ≥ 120) and heart disease was 46.0, while the percentage of those with fbs but no heart disease 46.5, indicating that fbs was not a relevant feature. After getting this initial sense, we performed a chi-square test and removed any feature with a p value less than 0.5. Once we performed this analysis, we cross-referenced our results with the "Machine Learning Classifications of Coronary Artery Diseases" [1], performed tests on different combinations of features, and came up with this set of features we would keep: cp, exang, slope, thal, trestbps, chol and thalach.
- For the penguin set, we computed the mean squared difference for each class and compared them pairwise, finding similar values. This suggested that the "sex" feature did not contribute significantly to distinguishing between species, so we removed it.

As previously mentioned, we split the heart dataset into 2 (categorical and numerical features) and then created another dataset with all mixed features. In order to get a better understanding of the potential outcome from each dataset, we examined the average correlation between features and the target variable on each dataset to determine which was most highly correlated to the target. Through this method, the mixed dataset showed strongest overall correlations, suggesting that both numerical and categorical features contribute valuable information and should be included.

Dataset Type	Average Correlation with Target
Numerical	0.2953
Categorical	0.3054
Mixed	0.3008

Tabel 1: Average correlation of different dataset types with the target variable for heart dataset.

Feature	Adelie vs Chinstrap	Adelie vs Gentoo	Chinstrap vs Gentoo
culmen_length_mm	100.1971	76.4592	1.6021
culmen_depth_mm	0.0054	11.2267	11.7234
flipper_length_mm	32.7274	736.1755	458.4637
body_mass_g	724.8938	1921751.6972	1847828.9956
sex_MALE	0.0000	0.00016	0.00016

Tabel 2: Mean squared difference for each feature between penguin species.

5 Results

In the Heart Disease dataset, we found through experiments that the best decision tree depth is 3, and the test set accuracy is 88.33%. When the depth is small, the model cannot fit the data well, and when the depth is too large, the model may overfit, resulting in a decrease in the test set accuracy.

We further compared the impact of different cost functions on the performance of decision trees. In the Heart Disease dataset, when the Gini index is used, the test set accuracy reaches 88.3%, which is better than information entropy (85.0%) and misclassification rate (81.7%).

In the Penguin dataset, as the tree depth increases, the accuracy rises rapidly and basically approaches 100%. In this dataset, the decision tree can easily distinguish the three varieties.

For different cost functions, Gini and Misclassification can both reach a test accuracy of about 100%, and Entropy is a little lower but close to 98.5% or more. Since the data itself has obvious feature discrimination, the decision tree depth does not need to be too large to obtain high accuracy.

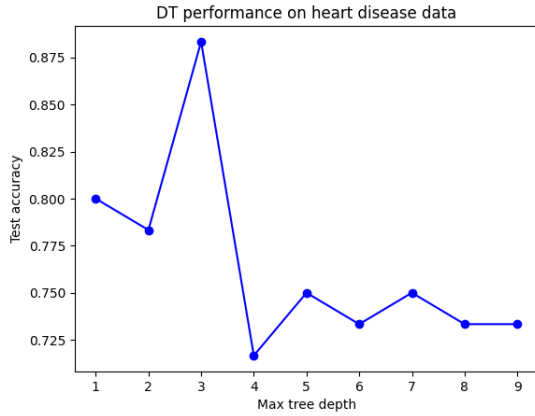
By analyzing the importance of features and the comparison of mean differences For complex data like heart disease, the results of the two methods are different because DT selects some features that can better classify the categories but have small mean differences. The mean difference method selects features with large mean differences but may not be the best for classification. For simple data like penguins, the results of the two methods are the same because the category characteristics of the data are obvious, the mean difference is large and can be well used for classification, so the DT method is more suitable for classification tasks, while the mean difference method is more suitable for analyzing the overall statistical characteristics of the data.

DT Importance	Mean Difference
heart_feat_12	heart_feat_7
heart_feat_11	heart_feat_4
heart_feat_2	heart_feat_3
heart_feat_0	heart_feat_0
heart_feat_3	heart_feat_12

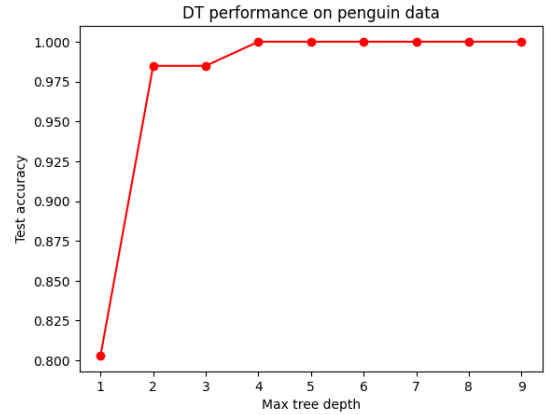
Tabel 3: Top 5 Features for Heart Dataset.

DT Importance	Mean Difference
culmen_length_mm	culmen_depth_mm
culmen_depth_mm	flipper_length_mm
flipper_length_mm	culmen_length_mm
body_mass_g	body_mass_g
sex_MALE	sex_MALE

Tabel 4: Top 5 Features for Penguin Dataset.



(a) Max depth of decision tree on Heart Disease dataset



(b) Max depth of decision tree on Penguin dataset

Figure 1: Comparison of decision tree depth for Heart Disease and Penguin datasets

For the Penguin dataset, K-Nearest Neighbors (KNN) performed really well, almost matching the accuracy of the Decision Tree model. Since the different species have pretty distinct physical traits, KNN had no trouble classifying them, even with minimal tuning. When testing different values of K, we noticed that after $K = 5$, the accuracy stayed about the same, showing that the species were already well-clustered. Using a smaller K (like 1 or 3) made the model more sensitive to noise, while a larger K (10 or more) smoothed things out but didn't really change the accuracy much. We also tried weighted KNN, where closer neighbors had more influence, but it didn't make much of a difference. The species were already easy to separate, so weighting wasn't needed. This suggests weighted KNN is more useful when class boundaries aren't as clear.

We also experimented with different distance metrics, Euclidean distance (default) worked great and was the most straightforward. As the datapoints were already well-separated other distance functions gave similar results and didn't increase the accuracies significantly.

Lastly, we looked at the ROC curves to compare KNN and Decision Trees in terms of classification confidence. Both models had near-perfect AUROC scores, confirming that this dataset was really easy to classify.

The heart dataset, in general, was a lot more difficult to classify than the penguin one, resulting in consistently worse scores overall. For mixed data the optimal k was 6, with a validation accuracy of 83, and test accuracy of 77. Only categorical data also performed well, and had a maximum test accuracy of 76. Finally, using only numerical data, the best test accuracy we found was with a k value of 4, and

was 61 percent. This confirms our previous prediction determined by the distribution of each dataset, which showed that the categorical and mixed datasets had significantly higher correlations to the target variable than the numerical dataset.

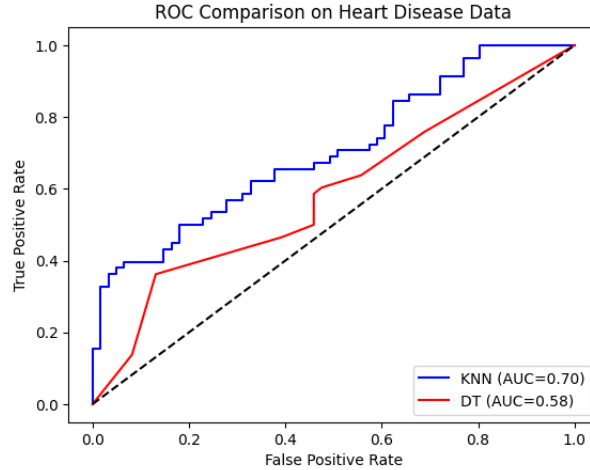


Figure 2: ROC comparison for KNN and DT for the heart dataset

6 Discussion and Conclusion

In the Heart Disease data, the accuracy fluctuates or even decreases when the depth is too large, indicating that some noise in the training set is overfitted. For the Penguin data, because the three breeds have significant differences in physical signs, it is easy to distinguish them at a shallow level, so the accuracy remains almost unchanged after the depth exceeds 4.

Gini, Entropy, and Misclassification can all obtain acceptable tree structures in binary classification, but Gini and Entropy are usually smoother and can better distinguish potential information gains. In our experiments, the Misclassification rate is not very sensitive, which will lead to some suboptimal node selection.

The decision tree implemented from scratch achieved quite good results on both data sets. The optimal test accuracy of the Heart Disease data is about 88%, while that of the Penguin data can reach 98%-100%. Different cost functions also have obvious differences in this data, and Gini performs relatively better in most cases. Regarding feature importance analysis, we found that the important features selected by the mean difference method and the decision tree method are not completely consistent. This may be because the mean difference method only considers the global mean change, while the decision tree calculates the importance of features based on the usage frequency of the split nodes.

For K-Nearest Neighbors (KNN) on the Penguin dataset, we found that it performed almost as well as the Decision Tree model. Since the three penguin species have pretty distinct physical traits, the classification was fairly easy, which explains that even with a simple distance-based approach like KNN we would get such great results. Testing different values of K showed that accuracy stayed consistent beyond a certain point, meaning the dataset was well clustered. We also tried weighted KNN, but it didn't make much of a difference, likely because the species were already easy to distinguish without extra weighting. Overall, this suggests that when a dataset has clear class boundaries, KNN can be just as effective as tree-based models.

With regards to the heart dataset for KNN, feature selection was very important when it came to optimising the performance of our model. When using the original data (with every feature present), the highest test accuracy was 64 for unweighted KNN and 75 for weighted KNN, both numbers that we improved by selecting features. In addition, using the mixed data yielded the best results.

7 Statement of Contributions

All team members contributed to the assignment, collaborating on the overall implementation, analysis, and report writing. Victoria focused on the Heart Disease dataset and implemented KNN for that dataset. Negar handled data processing for the Penguin dataset and developed the KNN model for it. Sam was responsible for implementing and optimizing the Decision Tree model.

8 References

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2. A. Shabani, M. Zandieh, S. F. Ghaderi, and S. M. Moattar, "The distance function effect on k-nearest neighbor classification for medical datasets," *SpringerPlus*, vol. 5, no. 1, pp. 1-13, 2016. Available: <https://doi.org/10.1186/s40064-016-2941-7>.
3. K. P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022. Available: <https://probml.github.io/pml-book/book1.html>.